

SENG 474 Data Mining - Assignment 3

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1 Exercise 1: Experiment and Analysis

1.1 Introduction

In this report, two clustering techniques are implemented and evaluated: Lloyd's algorithm (commonly known as k-means) and hierarchical agglomerative clustering. The implementation leverages Python, utilizing existing libraries such as Scikit-learn for hierarchical clustering, while Lloyd's algorithm is coded from the ground up. The analysis is performed on two distinct datasets: the first, dataset1, is a two-dimensional dataset comprising 3,500 samples generated from a Gaussian mixture model, and the second, dataset2, is a three-dimensional dataset with 14,801 samples, featuring intriguing underlying structures.

For Lloyd's algorithm, two initialization strategies are employed: uniform random initialization and k-means++ initialization. Experiments are conducted by testing various values of k (the number of clusters), running the algorithm multiple times for each k , and selecting the best outcome based on the sum of squared errors (SSE). The optimal number of clusters is identified using the elbow method, and the results are visualized to justify the chosen value of k . In the case of hierarchical agglomerative clustering, two linkage methods are examined: single linkage and average linkage. Dendrograms are utilized to determine the appropriate cut points for clustering, with detailed explanations provided for the selected thresholds. Finally, the performance of both methods is compared, and clear conclusions are drawn regarding their effectiveness and applicability to the given datasets.

1.2 The Data

As previously noted, the two clustering algorithms were tested on two distinct datasets. In this section, we aim to visualize the data to gain deeper insights into its structure. Visualization is a valuable step because clustering algorithms often perform differently depending on the characteristics of the dataset. First, we examine dataset1, as shown in Figure 1. A scatter plot is generated, with the color of each point determined by its index in the dataset. From this visualization, it appears that the data naturally forms four distinct clusters. This observation provides an initial hypothesis for the optimal number of clusters, which can be further validated during the clustering process. Moving on to dataset2, which is more complex due to its three-dimensional nature, we observe an intriguing structure. As illustrated in Figure 2, the data appears to curve and intertwine, resembling two planes coiled around each other. To highlight the underlying patterns, we created a 3D scatter plot where the color of each point corresponds to its

index, similar to the 2D plot. This coloring scheme allows us to clearly distinguish two primary clusters within the data. These visualizations not only enhance our understanding of the datasets but also provide a foundation for evaluating the performance of the clustering algorithms. By identifying potential cluster formations early on, we can better assess the effectiveness of the methods and refine our approach as needed.

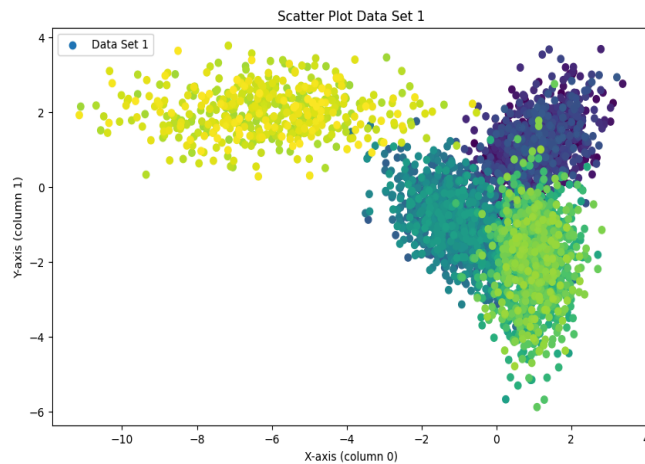


Figure 1: Scatter plot of Dataset1

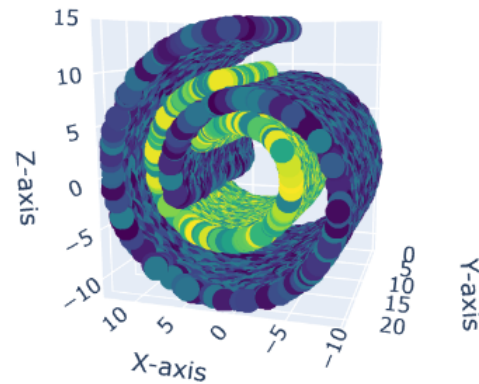


Figure 2: Scatter Plot of Dataset2

1.3 Lloyd's Algorithm:

Lloyd's algorithm, commonly referred to as k-means clustering, is an iterative method designed to partition a dataset into k clusters by minimizing the Sum of Squared Errors (SSE). The SSE quantifies the total squared distance between each data point and its assigned cluster centroid, serving as a measure of cluster compactness. The algorithm begins by initializing k centroids, either through random uniform initialization or a more sophisticated method like k-means++, which strategically places centroids to improve convergence. During the assignment step, each data point is assigned to the nearest centroid based on Euclidean distance, forming k clusters. In the update step, the centroids are recalculated as the mean of all points within each cluster. These steps repeat iteratively until convergence, defined as when the centroids no longer change significantly. However,

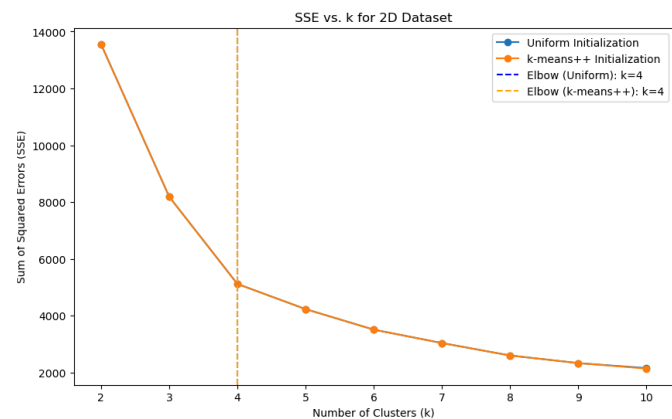


Figure 3: SSE vs k for dataset1

k-means is sensitive to initial centroid placement and can converge to local minima, necessitating multiple runs to identify the best clustering outcome.

In this exercise, we aimed to tune the hyperparameter k using the elbow method and silhouette score. The silhouette score evaluates how well each data point fits within its assigned cluster, providing insight into cluster cohesion and separation. For each value of k (ranging from 2 to 10), we ran Lloyd's algorithm multiple times, recording the lowest SSE values and calculating the silhouette score for the best clustering result. The elbow method was then applied to identify the optimal k by locating the point of maximum curvature in the SSE vs. k plot. For dataset1, the results, as shown in Figure 3, confirmed that the optimal number of clusters is four for both uniform and k-means++ initialization, aligning with our initial visual assessment. Figures 4 and 5 illustrate the four clusters, with both initialization methods achieving an SSE of 5117.4683 and a silhouette score of 0.493, indicating moderate clustering quality.

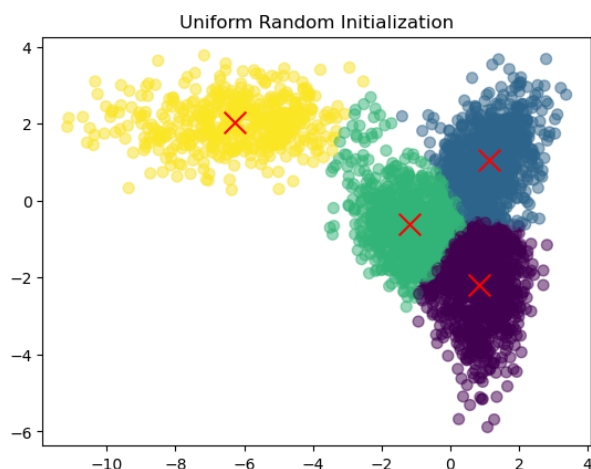


Figure 4: Uniform Initialization with optimal k on dataset1

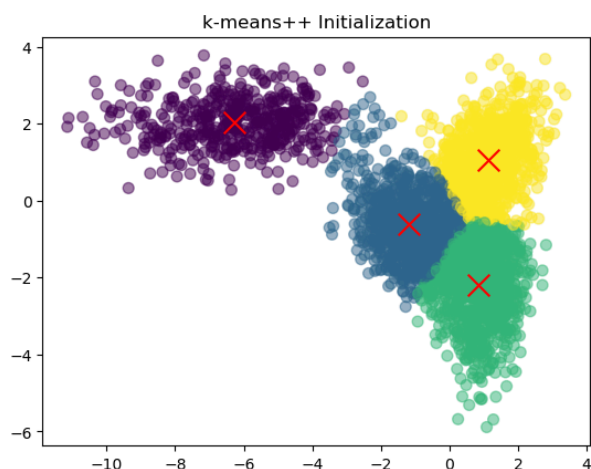


Figure 5: k-means++ Initialization with optimal k on dataset1

For the 3-dimensional dataset2, the elbow method suggested an optimal k of five, as depicted in Figure 6. However, upon visualizing the results, it became evident that this clustering did not fully capture the dataset's underlying structure. The SSE for both initialization methods was

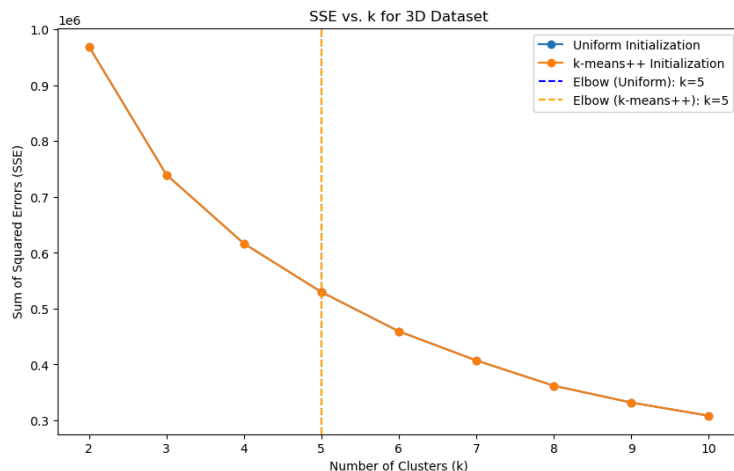


Figure 6: SSE vs k on 3-dimensional dataset2

529,867.922, and Figures 7 and 8 reveal that the clusters identified by k-means did not align with the expected two rolled planes. This discrepancy arises because Lloyd's algorithm assumes clusters are convex and roughly spherical, relying on Euclidean distance to assign points to centroids. As a result, it struggles with non-linearly separable data, such as intertwined or

manifold-like structures. In cases like dataset2, where clusters are curled around each other, k-means often imposes artificial boundaries that fail to reflect the true data distribution, highlighting a key limitation of the algorithm.

Despite its simplicity and efficiency, Lloyd's algorithm has notable limitations, particularly when dealing with complex or non-convex data structures. While the elbow method and silhouette score provide useful heuristics for selecting k , interpreting these metrics in

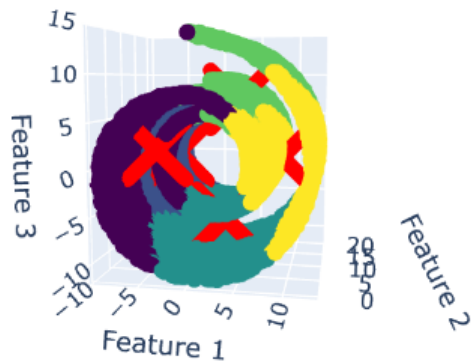


Figure 7: Uniform initialization with optimal k for dataset2

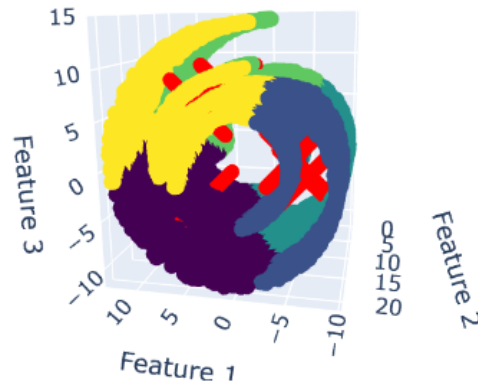


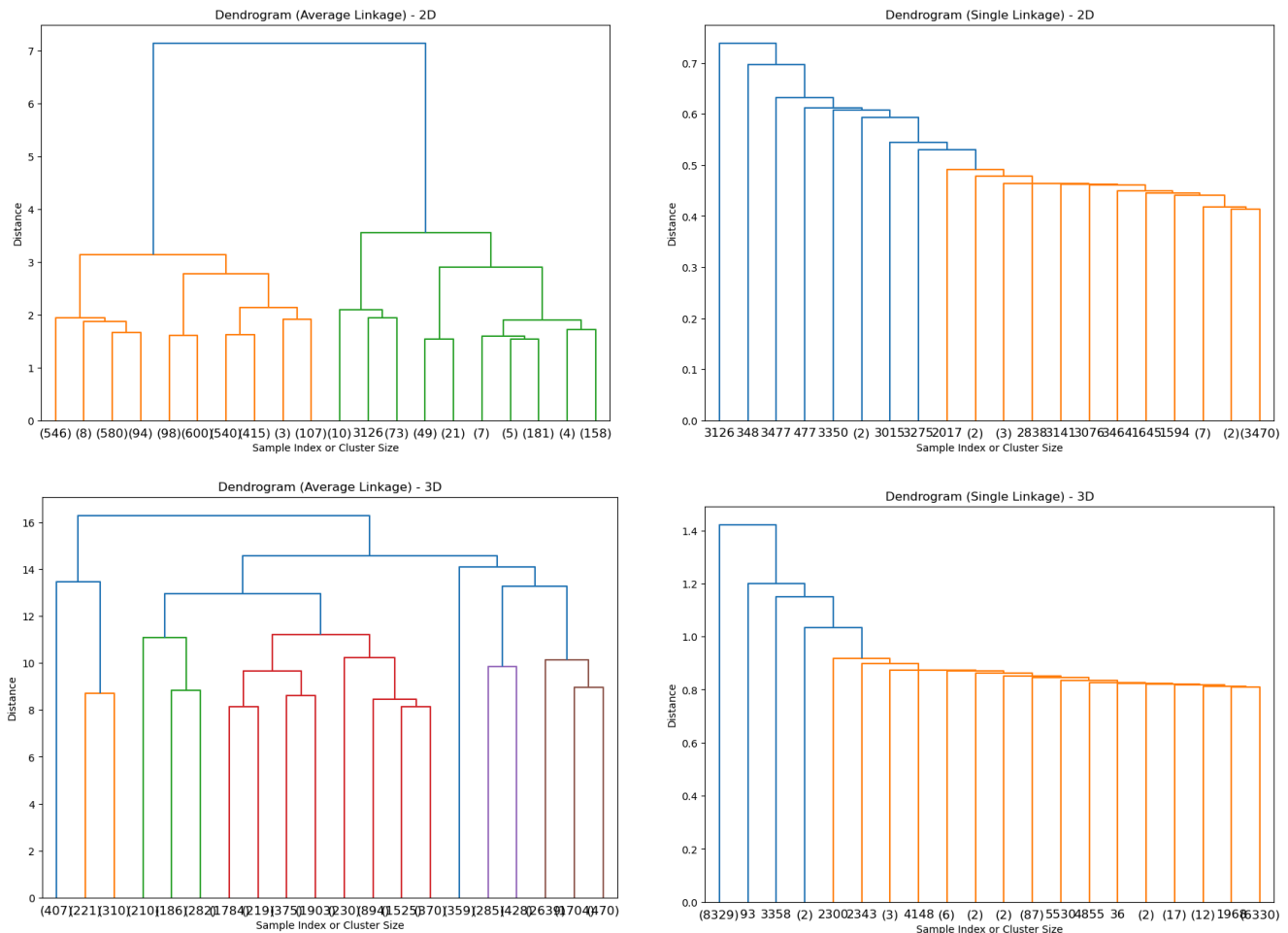
Figure 8: k-means++ initialization with optimal k on dataset2

higher-dimensional spaces can be challenging, as demonstrated by the 3D dataset. This underscores the importance of combining quantitative metrics with domain expertise and visual validation to ensure meaningful clustering results. Future work could explore alternative clustering techniques, such as hierarchical clustering, Gaussian mixture models, or density-based methods, which may better handle datasets with intricate or non-linear structures. By leveraging these approaches, we can achieve more accurate and interpretable clustering outcomes for a wider range of data types.

1.4 Hierarchical Agglomerative Clustering:

Hierarchical agglomerative clustering (HAC) is a bottom-up clustering technique that constructs a hierarchy of clusters by iteratively merging the most similar pairs of clusters. The algorithm begins by treating each data point as an individual cluster. At each step, it calculates pairwise distances between all clusters using a chosen distance metric (e.g., Euclidean distance) and a linkage criterion (e.g., single or average linkage) to determine cluster similarity. The two closest clusters are then merged, reducing the total number of clusters by one. This process continues until all data points are merged into a single cluster or until a predefined number of clusters is achieved. The result is a dendrogram, a tree-like diagram that visually represents the sequence of merges and the relationships between clusters at varying levels of granularity. This hierarchical structure allows for flexible analysis of cluster formations at different scales.

In this exercise, our goal was to determine the optimal number of clusters (k) by analyzing the dendrogram and employing a heuristic approach based on linkage distances. The dendrograms, shown in Figures 9–12, provide a visual representation of the clustering process. To identify the optimal k , we extracted the linkage distances from the hierarchical clustering linkage matrix, which records the distances at which clusters are merged. By computing the differences between consecutive distances, we identified the largest jump, which signifies the point where further merging of clusters becomes less meaningful. This jump serves as an indicator of the optimal k , as it represents the most significant change in cluster cohesion.



Figures 9-12: Dendrogram for single and average linkage on dataset1 and dataset2

For dataset1, the results of HAC were mixed. As seen in Figures 13 and 14, single linkage identified three clusters, but the outcome was suboptimal, forming one large cluster with two isolated points. This suggests that single linkage struggled to capture the true structure of the 2D data. Average linkage performed slightly better, identifying two main clusters, but overall, HAC did not yield particularly effective results for dataset1. In contrast, HAC demonstrated stronger performance on the 3D dataset2. As illustrated in Figures 15 and 16, average linkage

Clustering with Average Linkage ($k=2$)

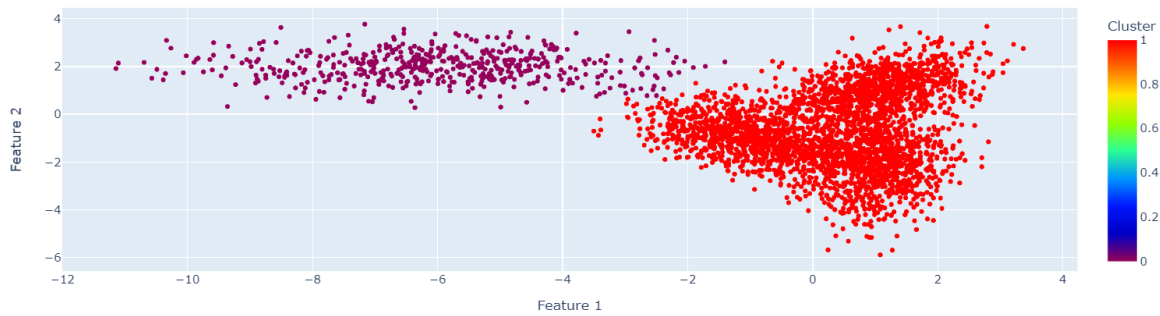


Figure 13: Average linkage on optimal k for dataset1

Clustering with Single Linkage ($k=3$)



Figure 14: Single linkage with optimal k on dataset2

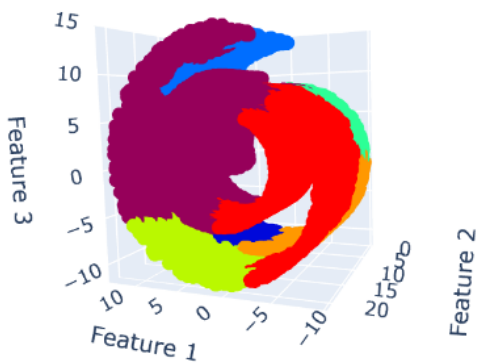


Figure 15: Average linkage with optimal k on dataset2

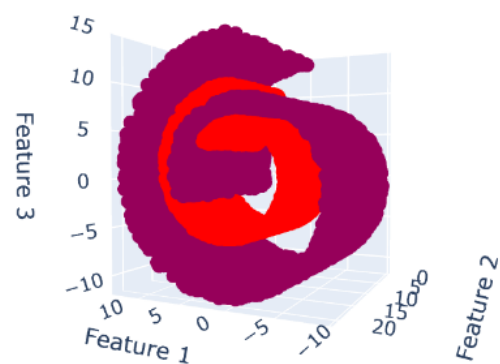


Figure 16: Single linkage with optimal k on dataset2

selected seven clusters, which did not align with the dataset's true structure. However, single

linkage successfully identified the two “rolled planes” as distinct clusters. This success can be attributed to HAC’s ability to handle non-convex and irregularly shaped clusters, unlike k-means, which assumes spherical clusters. Single linkage, in particular, excels at capturing elongated and intertwined structures by linking points based on their nearest neighbors, making it well-suited for complex datasets like dataset2.

Overall, hierarchical agglomerative clustering offers a flexible and powerful alternative to k-means, particularly for datasets with non-convex or irregular shapes. However, its effectiveness depends heavily on the choice of linkage criteria and the inherent structure of the data. While HAC struggled with the 2D dataset, it excelled in uncovering the true structure of the 3D dataset, underscoring the importance of selecting clustering methods that align with the data’s characteristics. Investigating other linkage criteria, such as complete or Ward linkage, could provide further insights into optimizing HAC for specific data types.

1.5 Comparative Analysis

A key takeaway from this study is that no single clustering algorithm universally outperforms the other; rather, their effectiveness depends on the structure and dimensionality of the data. The comparative performance of Lloyd’s algorithm (k-means) and hierarchical agglomerative clustering (HAC) varied significantly between the two datasets, illustrating the impact of data topology on clustering outcomes.

For the 2D dataset (dataset1), Lloyd’s algorithm proved to be the superior method. The Gaussian mixture structure of dataset1 aligns well with k-means’ assumptions of convex, well-separated clusters. The elbow method effectively identified four clusters, confirming the visual intuition derived from the scatter plot. K-means efficiently assigned points to their nearest centroids, minimizing the sum of squared errors (SSE) and yielding compact, well-defined clusters. HAC, in contrast, struggled with this dataset. Single linkage produced poor results by forming one large cluster with two isolated points, demonstrating its tendency to create elongated clusters that may not reflect the true distribution of the data. Average linkage performed slightly better but still failed to fully capture the expected four-cluster structure. These shortcomings highlight a key limitation of HAC: its reliance on hierarchical merging can be sensitive to noise and outliers, particularly in low-dimensional spaces where clusters are more clearly separated.

Conversely, hierarchical agglomerative clustering performed significantly better on the 3D dataset (dataset2). The complex, intertwined nature of dataset2—a structure resembling two rolled planes—posed a significant challenge for k-means. Because k-means relies on Euclidean distance and assumes spherical clusters, it struggled to partition the data meaningfully, as seen in the suboptimal clustering results where artificial boundaries were imposed. In contrast, HAC, particularly with single linkage, excelled in this scenario. Single linkage’s tendency to form chains of points allowed it to naturally detect the two intertwined sheets, effectively capturing the dataset’s true structure. This highlights one of HAC’s strengths: its ability to handle non-convex

clusters and datasets with complex topologies that cannot be adequately represented by simple centroid-based partitioning.

The fundamental difference between the two algorithms lies in their clustering assumptions. K-means partitions the data into clusters that minimize within-cluster variance, making it well-suited for compact, convex clusters such as those in dataset1. However, this rigidity makes it ineffective for data with irregular or non-linear separations, as seen in dataset2. HAC, on the other hand, does not impose a fixed cluster shape, making it more adaptable to non-convex structures. However, HAC can be computationally expensive, particularly for large datasets, and its effectiveness depends heavily on the choice of linkage criteria.

In summary, Lloyd's algorithm proved optimal for the 2D dataset due to its ability to efficiently separate convex clusters, while hierarchical agglomerative clustering, particularly with single linkage, demonstrated superior performance on the 3D dataset by identifying its non-linear structure. This comparison underscores the importance of selecting clustering techniques based on the specific properties of the data rather than relying on a single method for all scenarios. Future research could explore hybrid clustering approaches or employ additional techniques, such as DBSCAN or spectral clustering, to further enhance clustering performance across diverse datasets.

1.6 Conclusion

This report provided a comprehensive evaluation of two clustering techniques—Lloyd's algorithm (k-means) and hierarchical agglomerative clustering (HAC)—on datasets with distinct structural properties. Through rigorous experimentation, we analyzed the effectiveness of each method, highlighting their strengths and limitations.

Lloyd's algorithm demonstrated efficiency and robustness in clustering convex, well-separated clusters, as seen in dataset1, where the elbow method and silhouette scores validated the expected four-cluster structure. However, its reliance on Euclidean distance and spherical cluster assumptions rendered it ineffective for dataset2, where the clusters followed a non-linear, intertwined structure. Despite improvements offered by k-means++ initialization, the algorithm struggled to adapt to the dataset's true form, leading to misleading cluster assignments.

Hierarchical agglomerative clustering, on the other hand, exhibited greater flexibility, particularly in dataset2. While its performance on dataset1 was suboptimal—failing to clearly separate all clusters—it successfully captured the intricate structure of dataset2 using single linkage. Unlike k-means, which forces data into predefined spherical shapes, HAC's bottom-up merging strategy allows it to preserve the dataset's natural connectivity. This result underscores the importance of choosing clustering algorithms based on data topology rather than relying on a one-size-fits-all approach.

Ultimately, this study highlights the trade-offs inherent in clustering techniques. K-means remains a powerful tool for partitioning well-defined, convex clusters efficiently, while HAC provides deeper insights into complex, non-linearly separable structures. Future research could

explore hybrid clustering approaches that integrate the speed of k-means with the adaptability of HAC, or leverage advanced techniques such as DBSCAN or Gaussian mixture models to further refine clustering accuracy. By carefully selecting clustering strategies that align with data characteristics, we can improve interpretability and ensure more meaningful insights in real-world applications.